Q. 1 (Successes and failures)**.** a. $\{M > 8\}$ is the event where the first failure occurs after the 8_{th} trial. Since there are only two outcomes, *success and failure*, this is equivalent to getting only successes in the first 8 trials; that is

$$\{M > 8\} = \{8 \text{ consecutive successes}\}.$$

So

$$\mathbf{P}\{M > 8\} = \mathbf{P}\{8 \text{ consecutive successes}\} = p^8.$$

b. $\{M < N\}$ is the event that the first failure occurs before the first success. Since there are only two outcomes, the only way this is possible is if the first trial is a failure. Hence

$$\mathbf{P}\{M < N\} = \mathbf{P}\{\text{first trial is a failure}\} = 1 - p.$$

c. First note that for this probability to be non-zero either M or N has to be 1. So

$$\mathbf{P}\{M = i, N = j\} = \begin{cases} \mathbf{P}\{M = 1, N = j\} = \mathbf{P}\{N = j\} = (1-p)^{j-1}p \ \text{ for } i = 1 \text{ and } j = 2, 3, 4 \dots \\ \\ \mathbf{P}\{M = i, N = 1\} = \mathbf{P}\{M = i\} = p^{i-1}(1-p) \ \text{ for } j = 1 \text{ and } i = 2, 3, 4 \dots \\ \\ 0 \ \text{ otherwise.} \end{cases}$$

Q. 2 (To tweet or not to tweet?)**.** First, we define the events and random variables:
E : D.T. ordered the EggMcMuffin,
M : D.T. ordered the Oatmeal.
T : time to complete tweet (in hours),
B : time to cook breakfast (in hours).

We know $B|E = 1/6$, $B|M = 1/4$, $\mathbf{P}(E) = \mathbf{P}(M) = 1/2$, $T \sim exp(4)$.

a. We need to find $\mathbf{P}(T < B)$. Note that it is equivalent to use $<$ or $\leq$ since T is a continuous random variable.

Since E and M form a partition (of the entire sample space of possible breakfasts), by the law of total probability, we can write:

$$\begin{aligned} \mathbf{P}(T < B) &= \mathbf{P}(T < B|E)\mathbf{P}(E) + \mathbf{P}(T < B|M)\mathbf{P}(M) \\ &= \mathbf{P}(T < 1/6)\mathbf{P}(E) + \mathbf{P}(T < 1/4)\mathbf{P}(M) \\ &= (1 - e^{-4/6}) * 1/2 + (1 - e^{-4/4}) * 1/2 \\ &= 1 - 1/2 * (e^{-2/3} + e^{-1}) \end{aligned}$$

Notice in the second step we have used the values of B conditional on the two events E and M. On the 3rd step we have used the fact that if $X \sim exp(\lambda)$, then $\mathbf{P}(X < x) = 1 - e^{-\lambda x}$.

b. We are asked to compute $\mathbf{P}(M|T < B)$. We use Bayes Rule.

$$\begin{aligned}\mathbf{P}(M|T < B) &= \frac{\mathbf{P}(T < B|M)\mathbf{P}(M)}{\mathbf{P}(T < B)} \\ &= \frac{\mathbf{P}(T < 1/4)\mathbf{P}(M)}{\mathbf{P}(T < B)} \\ &= \frac{(1 - e^{-4/4}) * 1/2}{1 - 1/2 * (e^{-2/3} + e^{-1})} \\ &= \frac{1 - e^{-1}}{2 - e^{-2/3} - e^{-1}}.\end{aligned}$$

The denominator was found in part a.

Q. 3 (Exam)**.** a. Let N be your total score on the exam. Let N_i be your score on question i, $i = 1, ..., 10$. By linearity of expectation:

$$\mathbb{E}(N) = \mathbb{E}\left(\sum_{i=1}^{10} N_i\right) = \sum_{i=1}^{10} \mathbb{E}(N_i) = 10\mathbb{E}(N_1),$$

since the questions are identically distributed. By direct calculation

$$E(N_i) = 0.2 \cdot 0 + 0.2 \cdot 5 + 0.6 \cdot 10 = 7,$$

so the expected total score is 70.

b. Let M be your total wrong answers on the exam. Let M_i be equal to 1 if you got question i wrong, and 0 otherwise. By linearity of expectation:

$$\mathbb{E}(M) = \mathbb{E}\left(\sum_{i=1}^{10} M_i\right) = \sum_{i=1}^{10} \mathbb{E}(M_i) = 10\mathbb{E}(M_1),$$

since the questions are identically distributed. By direct calculation

$$E(M_i) = 0.2 \cdot 1 + 0.2 \cdot 0 + 0.6 \cdot 0 = 0.2,$$

so the expected total number of questions answered wrongly is 2.

c. let A_i be the event that you get question i wrong. Let B be the event that you have exactly one question wrong. Then:

$$\mathbb{P}(A_i|B) = \frac{\mathbb{P}(A_i \cap B)}{\mathbb{P}(B)} = \frac{0.2 \cdot 0.8^9}{\binom{10}{1} 0.2 \cdot 0.8^9} = \frac{1}{10}.$$

In the above, we use the fact that for B to happen, you have to get 9 questions correct and one question wrong which can be thought of as 9 successes and one failure. But you also have to consider the number of ways this can happen—that is you get one question wrong among 10 questions. This is given by $\binom{10}{1} = 10$. The event $A_i \cap B$ is the event that you get 9 questions right and the i_{th} question wrong. There is only one way this can happen.

Hence you are equally likely to have gotten any of the questions wrong; with a probability of 1/10 for each question.

Grading Comments

1. It was significantly easier to approach this problem by probabilistic reasoning rather than attack it with mathematical tools. The most common mistakes were:

   - part (a):
     - by far the most common issue here was that of students leaving their answer in the form $\mathbf{P}\{M > 8\} = 1 - \sum_{m=1}^{8} p^{m-1}(1-p)$. Since this is a geometric series, we expect you to use the geometric series formula which you've seen in class, treated in precept and used on your homework, to simplify this further.
     - modeling M as a Binomial random variable. By construction, M and N are geometric random variables with parameters $1-p$ and p respectively.
     - A few students wrote $\mathbf{P}\{M > 8\} = \mathbf{P}\{N \leq 8\}$ which is not true. $\{M > 8\}$ is the event that the first 8 trials are successes whereas $\{N \leq 8\}$ is the event that the first success occurs before the 8_{th} trial. Some wrote $\mathbf{P}\{M > 8\} = 1 - \mathbf{P}\{M < 8\}$ which is also not true.
   - part (b):
     - this was the hit or miss problem of this midterm: either a student got it right or got it wrong. Most students who got it right reasoned it out probabilistically. A few correctly used a conditioning approach. Most students who attempted conditioning however claimed that M and N were independent which is clearly not true (see next point).
   - part (c):
     - claiming M and N are independent and writing their joint distribution as a product of their marginal probability functions. This is not true because M and N are defined on the same process. Knowing one of them tells you something about the other. For example, if you know $M = 5$ then you automatically know that the first 4 trials were successes and hence $N = 1$.
     - Probably the most common mistake was forgetting to consider the third case: values of i and j where $\mathbf{P}\{M = i, N = j\} = 0$.
     - a few students recognized there was a third case but only discussed part of it. For instance some students noticed that $\mathbf{P}\{M = i, N = j\} = 0$ if $i = 1$ and $j = 1$ but this is incomplete. You also have to state the other possible cases where it is true. It is true for $i = j$ and $i, j \neq 1$.
     - some students combined the cases into one equation and wrote

       $$\mathbf{P}\{M = i, N = j\} = (1-p)^{j-1}p + p^{i-1}(1-p),$$

       which is wrong. For instance, if $i = 2$ and $j = 3$, this formula gives

       $$\mathbf{P}\{M = 2, N = 3\} = (1-p)^2 p + p(1-p) \neq 0.$$

       A few students did this approach with wrongly-defined indicator random variables.

2. Part b was answered well. For part a, the most common issues were the following.

   - Not converting to hours (or minutes) so that all the times are in the same unit (1 point taken off)
   - Some students created events/used notation in such a way that independence was needed to solve the question. If "by independence" was not explicitly stated, points were lost.
   - Just stating $\mathbf{P}(T < B|E)\mathbf{P}(E) = (1 - e^{-4/6}) * 1/2$ without any justification that we are using the exponential cumulative distribution function.

3. - An answer to the first two parts is not correct if it does not use linearity of expectation or the multinomial distribution. Finding 70 some other way is just educated guessing.
   - In the third part we need to be careful to use conditional probability. Knowing that $\mathbb{P}(A_i) = 0.2$ is not helpful.